

Samuel Xavier Borges

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Science, Minor in Data Science, Minor in Math | GPA: 3.5

2028

Orlando, FL

EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

Unary Lab - University of Central Florida

Orlando, FL

- Characterized the **Llama-2-7B** inference workload for the **Mugi LLM** accelerator by classifying all **2,500+** per-pass operations into four hardware execution buckets, then generating a validated **ONNX** graph consumed by the lab's cost/cycle simulator.
- Rearchitected the **Python/PyTorch** profiling pipeline to replace per-model if/else dispatch with a plugin adapter registry, uncovering and fixing a latent binning-format defect that had silently broken instrumentation for every model outside the one hardcoded configuration, unblocking profiling for additional architectures.
- Designed a one-shot **profiling API** producing per-layer lookup-table configs and hardware workload specs from a single config object, through a pluggable strategy/criterion sizer validated end-to-end against the simulator on an **NCSA Delta HPC** cluster, collapsing per-layer hand-editing into one command.
- Designed an in process **evaluation harness** that repatches lookup tables in place rather than reloading a **13GB** model per layer, converting a **32** round manual search into one unattended job that sweeps a 16-window grid at every layer, seeded per layer from measured exponent histograms.

Software Engineering Intern

Feb 2025 – Jan 2026

Knight Hacks

Orlando, FL

- Drove site **accessibility** to **WCAG 2.1 AA** keyboard navigation compliance by building **15 reusable React UI components** that replaced non-semantic SVG elements with properly labeled, **ARIA** attributed alternatives.
- Engineered an **email automation system** from scratch, delivering personalized acceptance, denial, and event-detail communications to **1,000+** applicants for one of **Florida's largest hackathons**.
- Built a database backed **email queue service** and **admin console**, adding a four tier priority scheduling, batch recipient validation, and date-range blacklisting to replace ad hoc one off send calls across the platform.

Software Engineering Intern

May 2025 – Aug 2025

Data-Enabled Photovoltaics

Orlando, FL

- **Automated parsing** and normalization of heterogeneous metadata across **five instrument sources** by developing an object-oriented **Python** parser with **Pandas** and **SQLite**, reducing manual preprocessing time by over **75%**.
- Defined and validated **FAIR** data model standards across **5** pipelines, integrating workflow outputs to preserve **metadata provenance** through cross pipeline handoffs and make downstream analyses reproducible from raw capture to database record.
- Unified **EXIF** and **TIFF** metadata from electroluminescence camera captures with binary Sinton flash-test records into one normalized **SQLite** schema, adding per file **failure logging** that surfaced malformed captures instead of silently dropping them.
- Documented the parser/outputter/database separation with an extension guide and **D2** architecture diagrams, giving non programmer researchers a path to onboard new instrument types without modifying existing pipeline code.

PROJECTS

Ctrl-ARM | *Python, Pandas, Electron, React, Google ADK, TensorFlow, Whisper*

- Built a body movement **accessibility** system that combined **EMG** signals with **IMU motion data** and a multimodal **Electron + React** overlay, achieving **1st at ShellHacks 2025's Microsoft AI Challenge**.
- Engineered **Pandas ETL** pipelines to ingest and transform 200Hz EMG/IMU streams, enabling per-user calibration, feature extraction, and consistent **~90%** classification accuracy across **12+** gestures.

Tariffix | *TypeScript, React, Vite, Tailwind CSS, MongoDB, Chrome Extension API, Gemini API, Cheerio*

- Enabled real-time tariff insights on **100K+** product listings via a **Vite-powered Chrome extension** backed by an **AI scraping pipeline** for metadata and HTS codes, winning **Bitcamp 2025's "Best use of MongoDB."**
- Achieved under **5% average error** on tariff cost estimates against official U.S. rates by comparing gathered data to a custom **MongoDB** dataset, automating precise calculations for end users through use of **AI agent networks**.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, Java, HTML, CSS, JavaScript, TypeScript, SQL

Libraries & Frameworks: React, Next.js, Node.js, Vite, Tailwind CSS, tRPC, PostgreSQL, SQLite, MongoDB, Pandas, numpy, PyTorch, ONNX, Electron, Google ADK

Developer Tools: Git, Unity, Vercel, Prisma, Drizzle, Figma, Linux, Docker, npnm, D2